



Fig. 2: Proposed solution for power management using programmable logic

for each supply is limited as the micro-controller cannot handle high voltages.

3. Additional discrete ICs are required for routine tasks of level-shifting, reset generation, hot-swap, etc.

4. Use of discrete components makes design less reliable and complex.

Integrated power measurement and control

Programmable logic devices (FPGAs and CPLDs) are known in the digital world for their inherent benefits like reduced design time, reprogramming and overall cost reduction. If programmable logic can accomplish board-level tasks like digital housekeeping, glue logic and power management, this can be an integrated power measurement and control function.

For example, power management solutions are available where analogue functions are integrated with a programmable logic device. These solutions are suitable for complex power measurement and control as they overcome the limitations mentioned in the previous approach.

The power manager device is a single integrated chip that can perform voltage monitoring, sequencing and control. Its in-built 10-bit ADC converts analogue voltage into digital. An integrated I²C interface allows communication with other devices. The power manager device can provide high-voltage outputs (HVOUs) and digital inputs/outputs – all of which can be

controlled via I²C. The power manager has an integrated programmable logic device that can be programmed by software to accomplish miscellaneous tasks. The programmed file can be downloaded in the device through JTAG.

The system can use an external CPLD that has sufficient number of logic cells (LUTs) to implement a flexible and user-specific power controller function. The CPLD with built-in embedded functional blocks for I²C, SPI and timers can be used to accomplish the miscellaneous board-level tasks mentioned above. It can be programmed by JTAG or through I²C/SPI. The basic concept of such a design is shown in Fig. 2.

The power manager measures the supply voltage and the amplified voltage dropped across the series resistor. The ADC converts these voltages into digital values and stores the values in a 10-bit ADC register. The ADC register can be accessed via the inbuilt I²C interface and the digital values communicated to an external I²C master on the CPLD. Multiple power managers can be connected via I²C bus, each having a unique slave address. The power controller inside the CPLD implements the multiplier, divider and display driver specific to the user requirements. Any power control scheme can be implemented in the CPLD, which can be optimised for any given application.

The advantages of this approach are:

1. For this application, the power manager device can measure up to twelve rails of voltages. So in a multiple rail design, scaling up the system without board re-spin is possible.

2. The power manager has I²C controlled GPIOs and inbuilt CPLD to implement user-defined logic. Additional discrete components are not desired as all the routine jobs of supply sequencing, reset generation and hot-swap actions can be performed by the power manager.

3. The user can re-program the power manager device, thus providing flexibility for changing the design.

4. The embedded functional block I²C/timer implemented in the CPLD leaves enough LUTs for the designer to have a complex power controller that can have multiple functions of multiplier, divider, PWM control, LCD, RTC, etc. The flexibility associated with CPLDs is useful to design the controller depending upon the end application.

5. There are readily available IP and reference designs for the power manager, multiplier, PWM control and miscellaneous functions which designers can use to quickly create power management solutions for their boards, thus reducing the design time and effort.

To sum up

An integrated programmable power manager device provides the perfect combination to implement power control mechanism for power reduction and optimisation. It has the advantages of scalability, flexibility and ease of development while maintaining the accuracy.

The readily available IPs and reference designs can significantly reduce design time. However, the system designer should use his own discretion for choosing the power measurement and control technique depending upon the application. ●

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